Assignment No. 1

1. Solving Linear Systems Using Different Methods

Objective:
Understand and implement methods for solving linear systems of equations.

Tasks:

1. Solve the following system using **substitution and elimination methods**:

$$2x+y-z=8$$
$$-3x-y+2z=-11$$
$$-2x+y+2z=-3$$

2. Verify the solution by substituting the values back into the equations.
3. Write a short explanation of which method (substitution or elimination) was easier and why.

2. Programming Assignment: Gaussian Elimination

Objective:
Implement an algorithm to solve linear systems using programming.

Tasks:

1. Write a program in **Python / C++ / Java** that solves a system of **3 linear equations with 3 variables** using **Gaussian Elimination**.
2. The program should:

- Accept coefficients as input
- Convert the system to **row echelon form**
- Perform **back substitution**
- Display the values of x, y, and z

3. Test the program with at least **3 different systems of equations**.

3. Matrix Representation Assignment

Objective:
Understand how linear equations are represented using matrices.

Tasks:

Given the system:

$$x+2y+3z=9$$
$$2x-y+z=8$$
$$3x+y+2z=10$$

1. Write the system in the form:

AX= B

2. Identify matrices **A, X, and B**.
3. Solve the system using **matrix inverse method**.
4. Explain why matrices are useful in computer algorithms.

4. Real-World CS Application

Objective:
Apply linear systems to a computer science scenario.

Problem:

A server processes three types of tasks: **A, B, and C**.

- Task A uses **2 CPU units**, **1 RAM unit**, **1 GPU unit**
- Task B uses **1 CPU**, **3 RAM**, **1 GPU**
- Task C uses **1 CPU**, **1 RAM**, **2 GPU**

Total resources used are:

- CPU = 10
- RAM = 11
- GPU = 9

1. Form a **system of linear equations**.
2. Solve for the number of tasks **A, B, and C** executed.
3. Explain how linear systems help in **resource allocation in computing systems**.

5. Conceptual Questions

1. What is a **linear system of equations**?
2. What are the possible types of solutions?
    - Unique solution
    - Infinite solutions
    - No solution
3. What is the difference between **row echelon form** and **reduced row echelon form**?

4. How are linear systems used in **machine learning, graphics, and networks**?

6. Problem – Network Packets

A router sends packets **A, B, C**.

Bandwidth usage:

- A = 2 MB
- B = 3 MB
- C = 1 MB

Total packets = 10

Total bandwidth = 20 MB

Find possible numbers of A, B, and C.

7. Problem – Data Storage

Two types of files are stored:

- Type A = 2 MB
- Type B = 5 MB

Total files = 7

Total storage used = 24 MB

Find the number of each file.

8. Problem – Software Project Time

Three developers work on tasks:

- Developer A works **2 hours per module**
- Developer B works **3 hours per module**
- Developer C works **1 hour per module**

Total modules completed = 9

Total hours used = 18

Find number of modules completed by each developer.